the following window function:

$$\Psi(r,z) = \frac{1}{2\pi r_{\rm jet}^2} \exp\left(-\frac{r^2}{2r_{\rm jet}^2}\right) \frac{|z|}{h_{\rm jet}^2} \ . \qquad (25)$$

The window function is set to zero outside the cylinder defined by $0 \leq |z| \leq h_{\rm jet}$ and $r \leq r_{\rm jet}$. It is also normalized so that the total injected energy in the cylinder sums up to $\epsilon_{\rm f}\dot{M}_{\rm BH}c^2\Delta t$. In computing the value of Ψ for a given zone we use the zone-center coordinates.

For runs with jet feedback, in resetting the control surface region we want to exclude zones whose centers lie within the jet injection cylinder (both in computing the amount of material to remove and in applying the reset procedure). To achieve this, we move the jet inlet (the base of the cylinder) away from the $z = 0$ plane to start at $z = \pm R_{\rm reset}$. We have performed tests with different inlet offsets, but in those cases we found that often instabilities form below the injection cylinder and create outflows on the control surface away from the cylinder. Our intention is for the control surface region to serve as an inflow boundary except for where it intersects the injection cylinder, so we keep the jet offset at $R_{\rm reset}$ in all these runs.

3.2. *Tests of Jet Feedback*

To test the jet injection model, we performed three-dimensional hydrodynamical simulations with the sink particle method and the jet model within a box of size 256 pc in the xy plane and 512 pc along the z-axis. The grid was initialized to have uniform density and pressure values resembling ones at the center of the Perseus cluster (Fabian et al. 2006). This is a crude approximation of a real cluster, but since the gravitational potential of the SMBH dominates at this scale, and the gas scale radius is much larger than the box, a uniform setup is justified. The electron density and pressure are taken to be $n_e = 0.05$ cm^{-3} and $P = 0.67$ keV cm^{-3}, and the gas is initially at rest. The ICM is treated as a fully ionized gas with an adiabatic index $\gamma = 5/3$. We use a point-mass gravitational potential corresponding to an SMBH mass of $10^9 M_\odot$ (we also do tests with various SMBH masses). With these values, the accretion radius is $r_{\rm B} = 2.28$ pc except for runs with varying SMBH masses. The jet axis is taken to lie in the z-direction.

Cooling is ignored for these tests, as the cooling time is larger than the time followed in the simulations. Additional consideration of microphysics processes in the ICM is left for later work. The JA runs use a uniform mesh, but in the other series, we use three levels of mesh refinement, with the finest level having a zone spacing Δx. For different resolutions, we keep the block structure the same, but vary the number of zones per block. The base grid contains $8 \times 8 \times 16$ blocks. Run parameters for these tests are shown in Table 3.

3.2.1. *Resolution Dependence*

Here we consider the effect of spatial resolution on our feedback model (shown via runs in the JA series). The notion of convergence is complicated for subgrid models: as we consider smaller resolution elements, the "physically appropriate" subgrid model may change, and in the limit of infinite resolution we should not need a subgrid model at all. Self-convergence of a code never formally implies convergence to the correct solution, but when a subgrid model is involved we must also consider whether the model is valid at the scales under consideration. Keeping these considerations in mind, here we consider self-convergence over the range of zone spacings we might typically encounter in galaxy or galaxy cluster simulations. These spacings are much larger than the accretion disk size, so we do not expect the expanded suite of physics needed for global accretion disk simulations to apply.

Using our BHL tests as guidance, we have thus fixed the control surface and reset radii with respect to different resolutions, so as to have a better comparison for the relationship between the Bondi radius and the control surface radius. Runs JA1, JA2, and JA3 have $r_{\rm B} > R_{\rm cs}$, while the opposite is true for run JA4.

The time evolution of the most well-resolved run (with a uniform grid having resolution of $\Delta x = 0.5$ pc) is shown in Figure 10. In this figure we show the yz slice of the gas density field at $x = 0$ for different times in the simulation. The jets propagate along the z-axis, forming low density channels while inflating a cocoon of entrained gas. The jet reaches the edge of the box quickly (in less than 50 kyr), and the simulation converges to a steady state where the jet resembles a nozzle with secondary shock structures along the propagation axis. The secondary shock structures become more complex over time.

The jet flow at a later time in the simulations with different resolutions is shown in Figure 11. In the figure we see similar flow structures, and in all cases the jets reach the boundary of the domain. However, the structure of the gas flow along the jet axis varies significantly with resolution. For better-resolved runs, the shocked region is smaller and low density is observed along the jet axis. The secondary shock structures remain in horizontal distances close to the jet, but occur further away from the accretor vertically, and they do not seem to affect the flow through the control surface. A shock diamond is also observed. For poorly resolved

Table 3. Table of test runs including jet feedback. In each case the ambient medium was set up to resemble the center of the Perseus cluster as described in the text.

Objective	Run Number	$M_{BH}/M_\odot$	Δx (pc)	$R_{\rm cs}/\Delta x$	$R_{\rm reset}/\Delta x$	$r_{\rm jet}/\Delta x$	$h_{\rm jet}/\Delta x$
Vary resolution of $r_{\rm B}$ on uniform grid	JA1	1.0×10^9	0.5	8	4	2	2
	JA2	1.0×10^9	1	8	4	2	2
	JA3	1.0×10^9	2	8	4	2	2
	JA4	1.0×10^9	4	8	4	2	2
Vary resolution of $r_{\rm jet}$	JB1	1.0×10^9	0.25	8	4	1	4
	JB2	1.0×10^9	0.25	8	4	2	4
	JB3	1.0×10^9	0.25	8	4	3	4
	JB4	1.0×10^9	0.25	8	4	4	4
	JB5	1.0×10^9	0.25	8	4	8	4
Vary resolution of $h_{\rm jet}$	JC1	1.0×10^9	0.25	8	4	2	2
	JC2 (Same as JB2)	1.0×10^9	0.25	8	4	2	4
	JC3	1.0×10^9	0.25	8	4	2	8
	JC4	1.0×10^9	0.25	8	4	2	16
Vary $r_{\rm B}$ relative to all	JD1	3.0×10^9	0.25	8	4	2	2
	JD2 (Same as JC1)	1.0×10^9	0.25	8	4	2	2
	JD3	3.0×10^8	0.25	8	4	2	2
	JD4	1.0×10^8	0.25	8	4	2	2
Test with α-Bondi accretion rate	J0	1.0×10^9	0.25	8	4	2	2

runs however, the jet forms a high density channel and a backflow closer to the accretor.

With varying resolution in the JA series, the quantity that varies is $r_{\rm B}/\Delta x$. Based on the jet structure we observe, we need $r_{\rm B} > 2R_{\rm cs}$ in order to have a converged jet flow structure (corresponding to a resolution of 0.5 pc for this black hole mass). Since our tests run on a uniform grid, we are constrained by computing power, but for later runs, we use an adaptive grid to reach a higher resolution of 0.25 pc.

To examine the jet structure in more detail, in Figure 12 we plot azimuthally-averaged gas profiles in slices perpendicular to the jet axis at a late time for the runs. Since the jet lies along the z-axis, for each plot we choose a set of 100 z values, compute the azimuthally averaged density, pressure, and velocity magnitude versus distance $r = \sqrt{x^2+y^2}$ between 100 pc and $L_{\rm box}/2$ from the jet axis for each z value, and then average the profiles over the z values. To avoid boundary effects closer to the reset radius and the domain boundary, we take only the z values larger than the reset radius and up to 250 pc. The left panels in Figure 12 show results for the JA runs, varying key radii together with resolution.

In the density profiles of the JA runs, we see a drop from the central value for all cases. The poorer the resolution, the larger the turn-over radius, which is consistent with Figure 11, where poor resolution runs have jets that cover a larger region. However, more poorly resolved runs also show a lower average density outside the jet. Most of the jet material is in pressure equilibrium with the ambient medium, with a drop towards the central area closer to the spine of the jet. The differences for larger radii among runs are negligible. The jet flow speed at its maximum also remains consistent, reaching over $\sim 10^4$ km s^{-1} and dropping more than two orders of magnitude for the gas farther away. However, well-resolved runs exhibit a more rapid drop of velocity, since cells around the jet edge are under-resolved.

Using the cumulative distribution $M(>e_k)$ of specific kinetic energies e_k, we can determine the mass of gas entrained by the jet. In Figure 13 we show this distribution for the different jet runs. In all these curves there exist three main components. At low e_k we have undisturbed material at or near rest. The step on the left (typically at or below 10^{14} erg g^{-1}) separates this gas from the entrained material, while the step on the right (around 10^{16} erg g^{-1}) marks the jet itself. The sharp drop to zero (above 10^{18} erg g^{-1}) marks the spine of the jet. The mass of entrained material is then the difference between the value of $M(>e_k)$ at the base of the left step and its value at the base of the right step.

The JA runs show convergence for $M(>e_k)$ for $\Delta x \leq$ 1 pc, with $\sim 200-300 M_\odot$ of entrained material, suggesting that $r_{\rm B} > R_{\rm cs}$ is needed for convergence of this statistic. The JB and JC runs show some sensitivity to the jet injection radius $r_{\rm jet}$, with convergence for

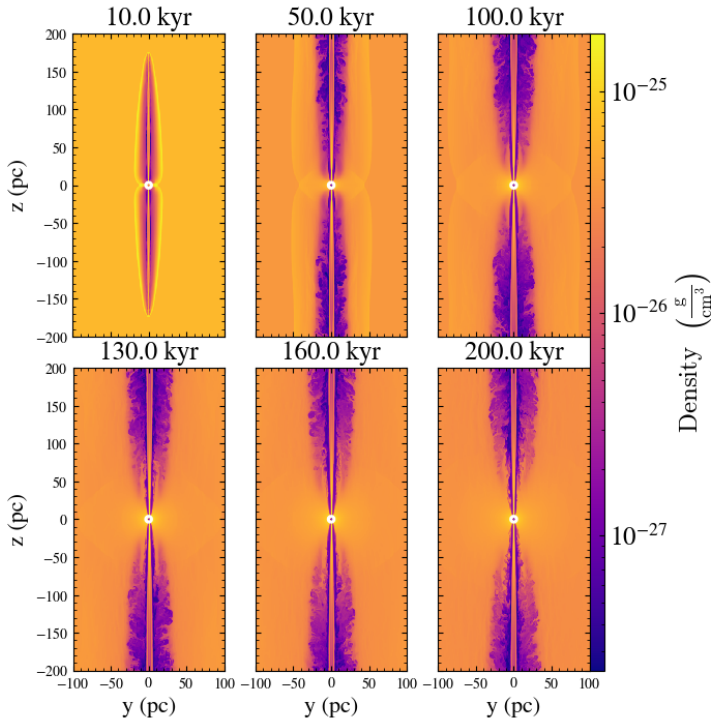

Figure 10. Density slices at different times for run JA1 with size of control surface $R_{\rm cs} = 4$ pc, fixing the jet size at $h_{\rm jet} = r_{\rm jet} = 2\Delta x$, where $\Delta x = 0.5$ pc. The white circle represents the control surface radius $R_{\rm cs}$.

$r_{\rm jet} \leq 2\Delta x$, but almost no dependence on the jet injection height $h_{\rm jet}$. The JD runs, which vary the Bondi radius relative to the other quantities, show a steady decrease in entrained mass with increasing $r_{\rm B}$, though the mass that is entrained moves faster. The lowest-mass run (JD4) does not resolve $r_{\rm B}$, and as a result $M(> e_k)$ shows some qualitative differences with the runs that do resolve $r_{\rm B}$.

The time evolution of the accretion rate, shocked volume, and enthalpy are shown in Figures 14 and 15. The accretion rate curve here is smoothed using a linear convolution with a sequence of length 50, i.e. smoothing every 50 data points. For visual inspection, the accretion rate is scaled to the Bondi-Hoyle-Lyttleton accretion rate. The shocked volume is calculated by summing up the volume of cells with a higher than ambient entropy $P/\rho^{5/3}$ in the domain. To allow for numerical errors, we set this threshold with a 3% error bar (which means if the entropy is higher than 97% of the ambient value, it is considered within the shocked region). The enthalpy is scaled relative to the ambient. For the latter two quantities, we do not include data inside the accretion radius or the jet inlet.

Despite accretion starting out differently, from $t \sim 50$ to 80 kyr onward, the accretion rate in most runs reaches a steady state where the accretion and feedback rate appear to stay relatively constant over time. For high resolution runs, the accretion rate stays around the Bondi-Hoyle-Lyttleton rate, but for lower resolutions, the accretion rate continues to increase, leading to an over-estimation. For the lowest resolution, however, there is a temporary under-estimation of accretion rate followed by accretion not able to reach steady state configuration. This is connected to the development of the jet feedback. Since fluctuations of the gas flow between different cells are better captured with increasing resolution, there is a more obvious self-regulation effect – more gas inflow through the control surface leads to stronger jet feedback, which pushes away more gas around the SMBH, and thus the accretion rate drops, leading to a fluctuation. For poorly resolved runs, despite the presence of the jet, it is also not able to regulate the inflow of the gas back onto the SMBH. In general, with resolution better than or equal to 1 pc, we are able to see the feedback transitioning into a more steady state configuration.

The volume of shocked material also increases as the jet cocoon expands, as seen in the middle panel of the figure. Eventually, the shocked volume fills up a large fraction of the grid. This value varies with resolution, as for the most well-resolved run, the shock only fills up about 10% of the grid, while for the run with resolution 2 pc, this number goes to about 60%, and 80% of the grid for poorest resolutions. This is partly due to the effect where, the simulation transitions into a more confined configuration for the jet for well-resolved runs. As turbulence increases within the jet region, there is more mixing of high-entropy and low-entropy material; thus some of the cells are not counted towards the shock because of the low entropy. After decreasing the entropy threshold (adding the 3 % error bar), the shock volume increases for all times in the simulation. Overall, the time evolution behavior shows consistency between shock structure and jet power, and further the accretion rate onto the SMBH. The enthalpy also demonstrates a behavior that is consistent with the above conclusions. While enthalpy increases initially a the jet is filling up the simulation domain, higher resolution runs show jets filling up quicker and the enthalpy decreases afterwards.